

Concept	Definition
Object-Oriented Programming (OOP)	Object-Oriented Programming (OOP) is a programming paradigm that organizes code into objects containing data and methods to improve reusability, maintainability, and scalability.
Object	An object is a collection of key-value pairs that represent data and behavior in JavaScript.
Class	A class is a blueprint used to create objects with predefined properties and methods.
Instance	An instance is an object created from a class using the <code>new</code> keyword.
Constructor	A constructor is a special method in a class that is automatically called when a new object is created to initialize its properties.
Method	A method is a function defined inside an object or class that performs a specific task.
Property	A property is a variable associated with an object that stores data.
Encapsulation	Encapsulation is the process of bundling data and methods into a single unit while restricting direct access to internal details.
Abstraction	Abstraction is the process of hiding implementation details and exposing only the essential features of an object.

Inheritance	Inheritance is the mechanism by which one class acquires the properties and methods of another class, promoting code reuse.
Polymorphism	Polymorphism is the ability of different objects or classes to provide different implementations of the same method.
Prototype	A prototype is an object from which other objects inherit properties and methods.
Prototype Inheritance	Prototype inheritance is the mechanism by which an object inherits properties and methods from another object's prototype.
Prototype Chain	The prototype chain is a sequence of linked objects that JavaScript follows to find properties and methods that are not directly available on an object.
extends Keyword	The <code>extends</code> keyword is used to create a child class that inherits from a parent class.
super Keyword	The <code>super</code> keyword is used to access the constructor or methods of a parent class from a child class.
Static Method	A static method is a method that belongs to the class itself rather than its instances and is called using the class name.
Static Property	A static property is a property that belongs to the class itself instead of individual objects created from the class.

Getter	A getter is a special method that retrieves the value of an object's property as if it were a normal property.
Setter	A setter is a special method that updates the value of an object's property while allowing validation or additional logic.
Private Field	A private field is a class property declared with # that can only be accessed within the class where it is defined.
Public Field	A public field is a class property that can be accessed from anywhere after an object is created.
Method Overriding	Method overriding is the process of redefining a parent class method in a child class with a different implementation.
Method Overloading	Method overloading is the ability to define multiple methods with the same name but different parameters. JavaScript does not support true method overloading; the last declared method replaces previous ones.
Composition	Composition is a design technique where objects are built by combining smaller, reusable objects instead of using inheritance.
Association	Association is a relationship in which two or more independent objects interact with each other.

Aggregation	Aggregation is a weak "has-a" relationship where one object contains another object, but both can exist independently.
Composition (Strong Relationship)	Composition is a strong "has-a" relationship where the contained object's lifecycle depends on the parent object.
this Keyword	The <code>this</code> keyword refers to the object that is currently executing the method or function.
new Keyword	The <code>new</code> keyword creates a new object from a constructor function or class and initializes it.
Constructor Function	A constructor function is a regular function used with the <code>new</code> keyword to create and initialize objects before ES6 classes were introduced.
Instance Method	An instance method is a method that belongs to an object created from a class and can access that object's properties.
Object Literal	An object literal is a simple way of creating an object using curly braces <code>{}</code> with properties and methods.
Object.create()	<code>Object.create()</code> creates a new object using an existing object as its prototype.
Object.freeze()	<code>Object.freeze()</code> makes an object immutable by preventing additions, deletions, and modifications of its properties.

Object.seal()

Object.seal() prevents adding or deleting properties but allows modification of existing properties.

Object.preventExtensions()

Object.preventExtensions() prevents new properties from being added to an object while allowing modification and deletion of existing properties.

Object.assign()

Object.assign() copies enumerable properties from one or more source objects to a target object.

Factory Function

A factory function is a function that creates and returns objects without using the `new` keyword or classes.